

Optimal Assimilation and Bias Correction Strategies for Surrogate Air Quality Forecasting in Urban Domains

The transition from deterministic Chemical Transport Models (CTMs) to statistical and machine-learning-based surrogate models for air quality forecasting necessitates a rigorous reevaluation of how observational data is assimilated into the modeling pipeline. In highly complex urban domains such as Greater Noida and the broader Delhi National Capital Region (NCR), which are characterized by extreme particulate matter (PM_{2.5}) concentrations, intricate urban topography, and highly localized emission sources, traditional data assimilation methods designed for prognostic physics engines often yield diminishing returns when ported directly to statistical architectures.

This comprehensive analysis evaluates the theoretical and practical utility of integrating observations from 40 Central Pollution Control Board (CPCB) surface monitoring stations within a 1,120-cell grid into a surrogate forecasting framework. The evaluation specifically addresses the assimilation of a coarse, 40-kilometer spatial resolution chemistry prior from the Copernicus Atmosphere Monitoring Service (CAMS), which characteristically exhibits a low concentration bias over the Delhi region during high-pollution episodes. The subsequent sections critically examine the standard formulations of optimal interpolation and successive correction, the mathematical redundancy of pre-correcting a chemistry prior, the operational cadences of state-of-the-art CTMs including the MoES AIRWISE Decision Support System, and the comparative efficacy of Kalman filter post-processing. The central thesis developed throughout this analysis demonstrates that for an algorithm already trained on localized station data, initializing a spatial state via classical assimilation offers marginal predictive value. Instead, lightweight post-processing via sequential filtering and native machine learning bias-mapping deliver substantially higher returns on computational investment.

The Mathematics and Application of Optimal Interpolation and Successive Correction

Optimal Interpolation (OI) and successive correction methods represent the foundational algorithms for blending a background model state (the prior) with observed data to produce a minimized-error analysis field. While these methods were originally conceptualized for synoptic-scale meteorology, they have been extensively adapted for surface air quality monitoring and the initialization of chemical transport models like the Community Multiscale Air Quality (CMAQ) model and the Weather Research and Forecasting model coupled with Chemistry (WRF-Chem)¹.

Standard Formulation of the Analysis State

The primary objective of an OI scheme is to compute an analysis state vector that optimally combines the background state vector and the observation vector to minimize the expected

analysis error covariance. The standard mathematical formulation for this update is expressed as an addition of an observation increment to the background state:

$$\mathbf{x}_a = \mathbf{x}_b + \mathbf{B}\mathbf{H}^T(\mathbf{H}\mathbf{B}\mathbf{H}^T + \mathbf{R})^{-1}(\mathbf{y} - \mathbf{H}\mathbf{x}_b)$$

In this foundational equation, $\mathbf{x}_a$ represents the analyzed concentration field, which in the context of the specified system, would be the updated PM2.5 concentrations across the 1,120-cell domain. The vector $\mathbf{x}_b$ represents the background concentration field provided by the 40-km CAMS prior. The vector $\mathbf{y}$ contains the actual surface observations from the 40 CPCB monitoring stations. The matrix $\mathbf{H}$ operates as the observation operator, interpolating the gridded model space to the specific geographical coordinates of the observations³.

The critical components governing the spatial influence of the observations are $\mathbf{B}$, the background error covariance (BEC) matrix, and $\mathbf{R}$, the observation error covariance matrix.

The matrix $\mathbf{R}$ accounts for both instrumental measurement error and representativeness error, which is the error arising from comparing a point measurement to a volumetric grid cell average¹. The term $(\mathbf{y} - \mathbf{H}\mathbf{x}_b)$ is defined as the innovation, representing the difference between the actual observation and the model's prediction at that exact location. The

weighting matrix applied to this innovation, $\mathbf{B}\mathbf{H}^T(\mathbf{H}\mathbf{B}\mathbf{H}^T + \mathbf{R})^{-1}$, is the Kalman gain matrix for a single, stationary assimilation step.

Successive correction methods simplify this computationally expensive matrix inversion through an iterative approach. Rather than solving the full optimal interpolation equations simultaneously, successive correction applies a simplified distance-dependent weighting function repeatedly across multiple passes. In each pass, the radius of influence is typically decreased, allowing the background field to be gradually pulled toward the observations, first capturing broad regional gradients and subsequently resolving finer, localized concentration peaks¹.

Background Error Covariance and Urban Length Scales

The efficacy of both optimal interpolation and three-dimensional variational (3D-Var) data assimilation is overwhelmingly dictated by the formulation of the Background Error Covariance

matrix, $\mathbf{B}$ ⁴. In standard optimal interpolation, the spatial correlation structure of the model prior is assumed to be homogeneous and isotropic, meaning the correlation function with respect to any reference point is invariant under translation and direction¹. The correlation

between the background error at two distinct geographic points separated by a distance r is parameterized by a horizontal length scale, L ¹.

Commonly utilized correlation models in air quality assimilation include the Gaussian decay function:

$$C(r) = \exp\left(-\frac{r^2}{2L^2}\right)$$

And the Balgovind, or first-order autoregressive, function:

$$C(r) = \left(1 + \frac{r}{L}\right) \exp\left(-\frac{r}{L}\right)$$

The specific determination of the horizontal length scale for urban pollutants is a highly complex process, typically requiring historical archives of observation-minus-model residuals. Methods such as the National Meteorological Center (NMC) lagged-forecast method or the Hollingsworth-Lönnberg approach are utilized to estimate these correlation lengths empirically by analyzing the spatial structure of forecast errors over extended periods¹.

Choosing the correct length scale for a domain like Delhi NCR is paramount. Unlike long-lived greenhouse gases or synoptic weather variables, PM_{2.5} exhibits a notably smaller horizontal length scale than most other atmospheric species⁷. This is because urban PM_{2.5} is dominated by highly localized anthropogenic emissions, such as dense vehicular traffic corridors, localized construction dust, and specific industrial or biomass-burning point sources, all of which are frequently trapped by shallow wintertime planetary boundary layer inversions⁹.

In regional or synoptic-scale background models, the horizontal length scale might be configured between 100 km and 300 km to spread corrections over broad geographical areas¹¹. However, in heavily urbanized micro-environments, applying such a large length scale is detrimental. If a large length scale of 50 km is erroneously applied in the Delhi domain, an observation from a highly localized traffic hotspot, such as the Anand Vihar monitoring station, will artificially inflate the analyzed concentrations in cleaner, upwind suburban grid cells, destroying the spatial heterogeneity of the model. Conversely, if the length scale is set too small, such as 1 km, the 40 CPCB stations will merely produce isolated "bullseye" corrections around the sensors, leaving the vast majority of the 1,120-cell domain entirely dependent on the biased, uncorrected CAMS prior¹².

Recent advancements in multi-scale three-dimensional variational assimilation (TS-3DVAR) have demonstrated that correctly constraining both large-scale background flows and small-scale urban variations is critical. Studies applying TS-3DVAR to dense monitoring networks have shown that resolving spatial wavelengths in the range of 54 km to 216 km yields significantly higher accuracy in the initial chemical field, reducing the root mean square error (RMSE) of PM_{2.5} analyses by up to 16.4% compared to standard 3D-Var approaches¹¹. For intra-city scales, empirical variograms frequently suggest that the optimal decorrelation length scales for PM_{2.5} reside on the order of 5 km to 20 km¹.

Assimilation Scale Context	Typical PM _{2.5} Length Scale (L)	Implications for Spatial Analysis
Regional / Synoptic Transport	100 km – 250 km	Generates broad, homogeneous adjustments; completely fails to resolve urban street canyons or

		localized emission spikes.
Urban Background / City-Wide	20 km – 50 km	Captures total city-wide pollutant accumulation; tends to smooth over localized industrial or traffic hotspots.
Traffic / Industrial Hotspot	2 km – 10 km	Produces highly localized corrections; requires an extremely dense network of low-cost sensors or reference stations to prevent spatial discontinuities.

Pre-Processing the Chemistry Prior: Bias Correction vs. Native Machine Learning

A foundational challenge in the specified forecasting architecture is the utilization of a chemistry prior from the Copernicus Atmosphere Monitoring Service (CAMS) at a coarse 40 km resolution, which is known to be biased low over the Delhi region. This low bias is a ubiquitous phenomenon in atmospheric modeling; global or coarse-resolution regional CTMs consistently underestimate wintertime PM_{2.5} concentrations in the Indo-Gangetic Plain. This underestimation stems from the inability of a 40-km grid to adequately parameterize shallow, dense planetary boundary layer inversions, combined with unresolved sub-grid emissions such as localized residential biomass burning and solid waste combustion⁹.

The architectural dilemma is whether it is materially advantageous to bias-correct this CAMS prior against the 40 CPCB stations through a pre-processing step (such as linear regression or optimal interpolation) before it enters the surrogate model as a feature, or to supply the raw, biased CAMS data and allow the machine learning algorithm to natively learn the necessary bias mapping.

The Mathematical Redundancy of Pre-Correction

In a statistical or machine-learning surrogate model, whether utilizing tree-based ensembles like Histogram-based Gradient Boosting (HGBost) or deep learning architectures like Convolutional Neural Networks (CNNs), the algorithm is designed to map a set of input

features $\mathbf{X}$ to a target variable $\mathbf{y}$ by minimizing a specified loss function¹⁶.

Assuming the input feature space $\mathbf{X}$ consists of the raw CAMS prior (x_c) and a rich set of

localized spatiotemporal and meteorological variables (x_m) such as boundary layer height, wind speed, relative humidity, and temporal embeddings. The surrogate model approximates a complex, high-dimensional function:

$$f(x_c, x_m) \approx y$$

If an offline bias correction procedure is applied to the prior to create a corrected feature $x'_c = x_c + \Delta(x_m)$, the surrogate model is subsequently tasked with learning a new function:

$$g(x'_c, x_m) \approx y$$

From an information theory perspective, the corrected variable x'_c contains no novel entropy or variance that is not already explicitly present in the joint distribution of (x_c, x_m) . If the bias $\Delta(x_m)$ is systematically driven by meteorological conditions—for example, the CAMS model underpredicts heavily when the boundary layer drops below 300 meters during January nights—a capable machine learning algorithm will naturally construct this exact relationship within its latent space or decision trees¹⁶.

Evidence from Surrogate Modeling Literature

Empirical evidence across the atmospheric data science literature consistently demonstrates that advanced machine learning architectures do not require pre-corrected inputs to achieve high predictive skill, provided that the variables driving the physical bias are included as features in the training dataset¹⁸.

The primary risk of applying a pre-correction step is the destruction of physically consistent multivariate relationships present in the raw CTM output. Chemical transport models generate outputs that are balanced according to physical and chemical laws. If CAMS provides both PM2.5 and aerosol optical depth (AOD), or precursors like NO2 and SO2, shifting the PM2.5 mean independently through a localized statistical correction disrupts the physical ratios between these variables²⁰. Surrogate models can exploit these raw, physically consistent relationships more effectively than altered, statistically disjointed ones.

Studies utilizing machine learning for PM2.5 forecasting demonstrate exceptional capability in native bias correction. For instance, the application of HGBost models utilizing raw pollutant

and meteorological data has achieved R^2 values exceeding 0.85 in testing phases, natively outperforming models that rely on simpler linear assumptions¹⁶. Similarly, hybrid CNN-Random Forest models have demonstrated the ability to extract key meteorological and pollution features without pre-processing, yielding RMSE improvements of 8.10% to 11.11% over standalone models¹⁷.

Furthermore, a study explicitly comparing the Analog Ensemble (AnEn) technique and Long Short-Term Memory (LSTM) networks for bias-calibrating CAMS forecasts found that these models could natively reduce the RMSE of the raw CAMS output by approximately 50% for

PM_{2.5} and 60% for PM₁₀¹⁵. The models inherently learned that the CAMS forecasts were underestimating PM concentrations by a factor of 2 during winter months due to misrepresented anthropogenic particulate emissions, and successfully scaled the predictions without any requirement for an intermediate, manual bias-correction step¹⁵.

The evidence dictates that it is not materially better to pre-correct the chemistry prior against the stations before it becomes a model feature. Supplying the raw CAMS data directly to the surrogate, alongside high-resolution localized meteorological features, prevents the introduction of statistical artifacts and eliminates an unnecessary computational bottleneck, allowing the loss-minimization objective of the algorithm to shape the highly non-linear, flow-dependent corrections natively¹⁵.

Operational Assimilation Cadences: CAMS Regional and MoES DSS

To contextualize what constitutes meaningful assimilation, it is imperative to analyze how operational, physics-based forecasting systems handle surface observations. These systems rely on prognostic equations of motion, mass conservation, and chemical kinetics, making their requirement for a precise initial state vastly different from that of a statistical surrogate model.

The CAMS Regional Production Architecture

The Copernicus Atmosphere Monitoring Service (CAMS) regional production is not a single deterministic model, but rather relies on a multi-model ensemble approach utilizing seven individual state-of-the-art European chemical transport systems²¹.

The data assimilation framework within CAMS is fundamentally reliant on advanced variational methods, predominantly 3D-Var and 4D-Var. The operational cadence is highly structured: CAMS-Global and Regional systems typically execute full forecast runs twice daily, initialized at 00:00 UTC and 12:00 UTC²². The assimilation window operates continuously leading up to these initialization times. Observations, including a vast network of surface monitoring stations and remote sensing satellite retrievals, are assimilated to update the background state, producing analyses every few hours. The resulting output data for surface fields is made available at an hourly cadence, providing a highly resolved temporal evolution of the assimilated state²².

The MoES Air Quality Warning and Integrated Decision Support System (AIRWISE/DSS)

For the specific geographical domain of Delhi and the surrounding NCR, the Ministry of Earth Sciences (MoES), in collaboration with the Indian Institute of Tropical Meteorology (IITM) and the India Meteorological Department (IMD), developed a highly specialized, operational forecasting framework known as the Air Quality Early Warning System (AQEWS), which was subsequently upgraded to the Air Quality Warning and Integrated Decision Support System for Emissions (AIRWISE)⁹.

The architectural foundation of the DSS is the Weather Research and Forecasting model coupled with Chemistry (WRF-Chem). To capture the extreme spatial heterogeneity of Delhi's urban canopy, the model utilizes a nested domain configuration, achieving an ultra-high spatial

resolution of 400 m × 400 m over the core city, downscaling from regional domains utilizing the EDGAR-HTAP v2.2 and high-resolution SAFAR emission inventories⁹. The DSS is uniquely designed to track anthropogenic PM_{2.5} tracers across 20 distinct districts within the NCR, allowing for dynamic source attribution⁹.

The assimilation mechanism employed by the AIRWISE framework is a Gridpoint Statistical Interpolation (GSI)-based 3D-Var and Ensemble Kalman Filter (EnKF) hybrid system. The system ingests and assimilates a massive volume of observational data, including near-real-time satellite Aerosol Optical Depth (AOD) retrievals from MODIS, active fire count data from the Visible Infrared Imaging Radiometer Suite (VIIRS) to account for regional stubble burning, and high-frequency surface PM_{2.5} data from approximately 320 monitoring stations operated by the CPCB, the Delhi Pollution Control Committee (DPCC), and IITM²³.

The operational cadence of the MoES DSS involves executing chemical data assimilation in cycles—often 3-hourly or 6-hourly, dictated by satellite overpass times and ground station telemetry latency—to generate a continuously optimized initial chemical condition²³. Based on this optimized initial state, the system executes a 72-hour forecast each morning. The implementation of this rigorous chemical data assimilation has been proven to significantly improve real-time forecasts, reducing the RMSE for Day-1, Day-2, and Day-3 forecasts considerably, and allowing the system to predict "Very Poor" air quality episodes up to 3 days in advance with a remarkable 83% categorical accuracy²³.

In these operational CTMs, spatial data assimilation is highly meaningful and strictly necessary. The updated initial state is physically advected by wind vectors, diffused by turbulence parameterizations, and chemically reacted over time by the prognostic Navier-Stokes equations and chemical mechanisms within WRF-Chem²⁷.

The Ephemerality of the Initial State vs. Kalman Filter Post-Processing

When operating a purely statistical or surrogate machine learning model—where atmospheric physics equations are replaced by static learned weights and activation functions—the utility of classical spatial assimilation changes drastically. The critical determining factor is the physical "memory" of the atmosphere versus the mathematical architecture of the surrogate model.

The Transience of Initial State Assimilation and e-Folding Time

In atmospheric chemistry modeling, the influence of an assimilated initial condition on a forecast does not persist indefinitely; it decays rapidly over time. This rate of decay is quantified as the e-folding time, or the half-life of the forecast memory²⁸.

Unlike stratospheric dynamics or deep oceanography where initial states dictate behavior for weeks, surface aerosol concentrations are highly dynamic. They are governed by continuous surface emission fluxes, rapid boundary layer mixing, deposition, and synoptic ventilation³⁰. Empirical studies assimilating PM_{2.5} into deterministic models like WRF-Chem or CMAQ consistently demonstrate that the positive impact of modifying the initial state is highly transient. Using an Ensemble Optimal Interpolation (EnOI) or EnKF approach, the assimilation effect is most remarkable only in the first 12 hours of the forecast³². After 24 to 48 hours, the

forecast error converges back to the baseline, unassimilated model error, because the continuous influx of boundary conditions and emission inventories rapidly overwhelm the memory of the initialized state²⁶.

In a surrogate machine learning model that entirely lacks a prognostic physics engine to advect mass or simulate deposition, updating a 2D grid's "initial state" via Optimal Interpolation is a

mathematically inert operation. A surrogate model evaluates a matrix of inputs at time t to

predict concentrations at time $t + n$. If the ML model is already natively trained directly on the CPCB stations, interpolating those exact stations into a 2D map simply to feed the smoothed map back into the model as an input feature is an exercise in circular logic. It provides minimal predictive advantage while introducing the smoothing errors inherent in the Gaussian decay functions of the interpolation scheme¹².

The Superiority of Kalman Filter Post-Processing

Conversely, applying a Kalman filter (KF) as a post-processing bias correction step on the forecast output is a highly dynamic, computationally lightweight, and remarkably effective strategy for statistical and deterministic models alike¹⁹.

Instead of attempting to update a spatial grid prior to the forecast, the KF predictor updates the time-series bias at each specific monitoring location after the forecast is generated. The post-processing framework assumes that the forecast error possesses a systematic, time-evolving component. The state vector in this specific KF formulation represents the true bias of the forecast, which is updated sequentially through recursive equations¹⁹.

1. **The Prediction Step:** The bias for the current time step is projected forward based on the state estimate of the previous step, maintaining the stochastic characteristics of the environmental system.

2. **The Update Step:** When the actual station observation telemetry arrives, the true

forecast error (the innovation) is calculated. The Kalman gain (K) dynamically weights this new error against the historic error covariance. The correction at each time step is proportional to the innovation, moving the prediction toward the observation by the

fraction determined by K ¹⁹.

Empirical evidence strongly supports the deployment of KF post-processing for PM2.5 forecasting:

- A KF post-processing framework implemented in the AIRPACT air quality forecast system for the Pacific Northwest demonstrated massive improvements, reducing the mean daily bias from approximately -50% to $\pm 6\%$ on a monthly averaged basis, and cutting the absolute errors to typically less than 20%³⁶.
- When applied directly to machine learning predictions, such as Random Forest or XGBoost models operating at street-level sites, Extended Kalman Filter (EKF) post-processing acts as a highly effective online residual correction mechanism. The EKF evolves the error covariance dynamically, tracking evolving pollutant concentrations over time far better than static offline models¹⁹.

- In environments subject to rapid regime shifts and high non-stationarity—such as the sudden onset of upwind stubble burning in Punjab impacting the Delhi NCR—offline machine learning models inevitably suffer from concept drift. The KF dynamically tracks and corrects this drift in real-time. Studies utilizing operational KF post-processing on PM2.5 forecasts routinely demonstrate RMSE reductions of 15% to 40%, particularly during extreme pollution episodes or seasonal transitions³⁶.

For a surrogate forecasting system focused primarily on station-level prediction accuracy, KF post-processing represents a targeted, mathematically sound intervention. It acknowledges that the surrogate model's weights are static upon deployment, and adds a stochastic, online correction layer that tracks unmeasured real-time variances and sudden emission anomalies¹⁹.

Strategic Recommendations and Implementation

Hierarchy

Based on the synthesis of operational air quality assimilation theory, the specific atmospheric dynamics governing the Delhi NCR domain, and the mathematical properties inherent to surrogate machine learning modeling, the following provides a definitive statement on the utility of assimilation, followed by a ranked recommendation of methodologies to implement to achieve the highest reduction in forecast RMSE.

The Utility of Spatial Assimilation in Surrogate Models

Before ranking implementations, an absolute scientific reality must be established regarding the architecture of the proposed system: **If a statistical or machine learning surrogate model is already natively trained on the 40 CPCB station observations—utilizing them as primary training targets and historical lag features—classical spatial data assimilation (such as Optimal Interpolation or 3D-Var) to create an initial analysis state provides negligible predictive value.**

A surrogate model trained on point data acts as an ultimate minimizer of localized error at those specific geographic coordinates. Optimal Interpolation, by definition, merely smears the station data across unmonitored grid cells using a distance-decay function, attempting to guess the concentrations in the void spaces¹. If the surrogate's objective is to forecast future concentrations at the station locations themselves, feeding it an OI-interpolated state is redundant; the model already has access to the exact station data. If the goal is to forecast in the unmonitored cells, the machine learning model should simply predict directly on those cells using the raw CAMS prior, localized meteorology, and spatial coordinates, allowing the trained weights to infer the complex spatial gradients based on historical patterns. The traditional "assimilation analysis cycle" is a necessary artifact of prognostic physics models, but is conceptually misaligned and computationally wasteful for purely statistical forecasting.

Ranked Implementation Recommendations

Rank	Implementation Strategy	Mechanism of Action	Expected RMSE Improvement
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1	Kalman-Filter Post-Processing	Sequential online residual correction on the forecast output at each station.	15% to 40% [cite: 36, 37, 38]
2	Autoregressive Station Features	Direct integration of lagged station observations into the ML feature space.	20% to 50% (short-term) ¹⁵
3	Native Bias Learning from Raw Prior	Supplying the raw, biased CAMS prior directly to the surrogate without offline pre-correction.	8% to 12% over linear corrections ¹⁷
4	Optimal Interpolation (2D Initial State)	Smearing station data across the grid prior to inference.	< 5% (often negative at stations) ¹²

Rank 1: Kalman-Filter Post-Processing (Online Residual Correction)

The highest-value mathematical intervention is the implementation of a sequential Kalman Filter, or an Extended/Fractional Kalman Filter, applied directly to the output predictions of the surrogate model at each of the 40 CPCB stations. As detailed previously, this acts as a lightweight online learning mechanism. It instantly corrects for sudden, unmodeled emission events—such as acute biomass burning influxes or localized, severe temperature inversions—that the coarse CAMS prior and the static offline ML weights fail to anticipate. It tracks the real-time drift of the model bias and adjusts the subsequent forecast hours accordingly, requiring minimal computational overhead compared to spatial interpolation¹⁹.

Rank 2: Integration of Lagged Station Observations as Direct ML Features (Persistence)

Rather than constructing a complex assimilation cycle, the system must ensure the surrogate model explicitly receives the temporal history of the 40 stations (e.g., $PM_{2.5,t-1}$, $PM_{2.5,t-24}$, and rolling 6-hour temporal means) as direct input features. PM_{2.5} concentrations exhibit massive temporal autocorrelation. Advanced algorithms like LSTMs or HGBost will naturally weigh recent station observations far more heavily than a coarse CAMS prior for short-term (1 to 12 hour) predictions¹⁵. Incorporating these autoregressive features

typically yields massive improvements in short-term forecast horizons, effectively achieving "assimilation in the latent space" without the rigid constraints and spatial assumptions of a Background Error Covariance matrix¹⁵.

Rank 3: Let the Surrogate Learn the CAMS Bias Natively

The system should pass the biased ~40 km CAMS chemistry prior directly into the surrogate model alongside high-resolution local meteorological data (PBL height, wind speed, relative humidity) and temporal embeddings (hour of day, month, day of week). Tree-based ensembles and neural networks are universal function approximators. If the CAMS prior consistently exhibits a low bias by a factor of 2 during specific winter meteorological setups, the surrogate will organically learn to apply the necessary non-linear scaling multipliers natively during inference¹⁵. Manual pre-bias correction is a relic of less capable statistical methods and risks destroying the physical consistency of the CTM outputs²⁰.

Rank 4: Optimal Interpolation (OI) for a 2D Initial State

Running a classic 2D-OI algorithm to merge the 40 CPCB stations with the CAMS prior to create a fully initialized 1,120-cell grid before running the surrogate model is the lowest-value implementation. For station-level forecasting accuracy, the expected improvement is near zero. While it might provide a marginal spatial smoothing benefit if the surrogate architecture is purely a Convolutional Neural Network (CNN) that strictly requires dense, continuous

image-like feature maps¹⁷, it requires the length scale L to be meticulously calibrated to urban realities (e.g., $L \approx 5 - 10$ km). Because spatial assimilation is fundamentally designed to provide mass to a physics engine for advection, smearing localized station data across a grid via isotropic Gaussian decay introduces non-physical spatial assumptions that the surrogate model must then actively work to un-learn. It is a computationally expensive, high-effort, and fundamentally mismatched technique for a statistical modeling architecture.

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